

About Dataset

Content :

This data about previous cases of health insurance of charges, to know the charges of the following cases .

Columns

- age: age of primary beneficiary
- sex: insurance contractor gender, female, male
- Bmi: Body mass index, providing an understanding of body, weights that are relatively high or low relative to height, objective index of body weight (kg / m ^ 2) using the ratio of height to weight, ideally 18.5 to 24.9
- children: Number of children covered by health insurance / Number of dependents
- smoker: Smoking
- region: the beneficiary's residential area in the US, northeast, southeast, southwest, northwest.
- charges: Individual medical costs billed by health insurance

```
# Summary statistics for info
print("\n📊 Summary statistics:")
display(df.info())
```

```
📊 Summary statistics:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1496 entries, 0 to 1495
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1496 non-null   int64
1   sex         1496 non-null   object
2   bmi         1481 non-null   float64
3   children    1492 non-null   float64
4   smoker      1467 non-null   object
5   region      1479 non-null   object
6   charges     1496 non-null   float64
dtypes: float64(3), int64(1), object(3)
memory usage: 81.9+ KB

None
```

```
# Step 3: Nulls & Duplicates Check
print("\n🔍 Null values ratio per column:")
print(df.isnull().mean()* 100)

print("\n🔥 Number of duplicated rows:")
print(df.duplicated().sum())
```

```
🔍 Null values ratio per column:
age      0.000000
sex      0.000000
bmi      1.002674
children 0.267380
smoker   1.938503
region   1.136364
charges  0.000000
dtype: float64

🔥 Number of duplicated rows:
141
```

